

Leakage-audited multi-ion insertion-electrode learning workflow

1. Multi-ion data extraction

Li 2,774 | Na 416 | K 164



2. Battery-property dependency graph

Machine-readable target-feature relationships



3. Automatic leakage compiler

Target-specific P0-P4 descriptor protocols



4. Leakage-aware validation

Random, framework, family, chemical-system and ion holdouts



5. Physics-constrained multi-task learning

Voltage, capacity, energy, volume and stability



6. Uncertainty and applicability domain

Conformal intervals, tree spread and distance flags



7. Sodium candidate triage

Physics filters, transfer score and rank robustness



8. Quantum ESPRESSO spot-check

Fixed-geometry PBE 3.203 V; $|\Delta V|$ to denser reference 0.00036 V